Experimentation & Evaluation

Gabriele Bavota
Giuseppe Crupi



Lecture 07

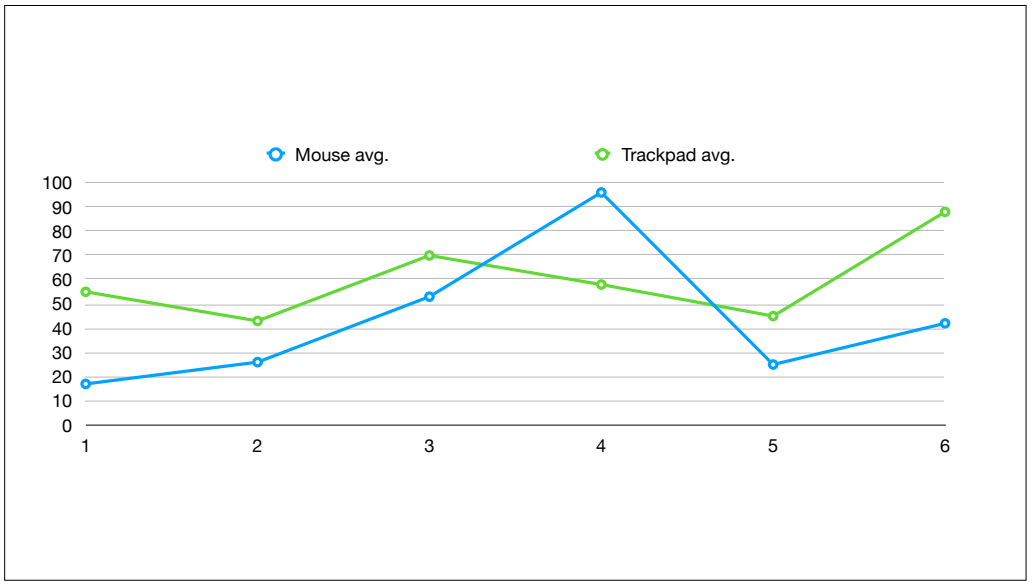
Confidence Intervals

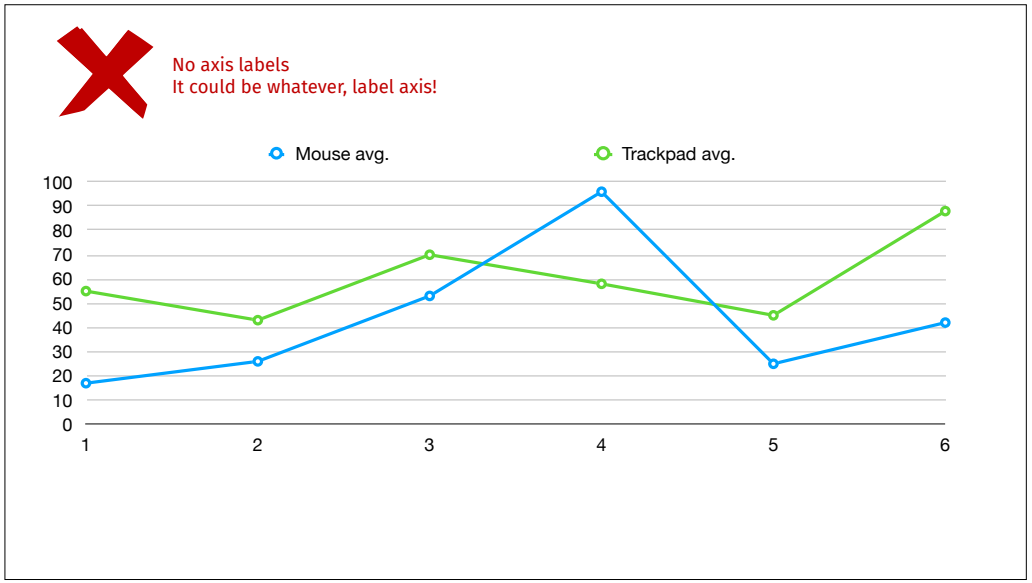


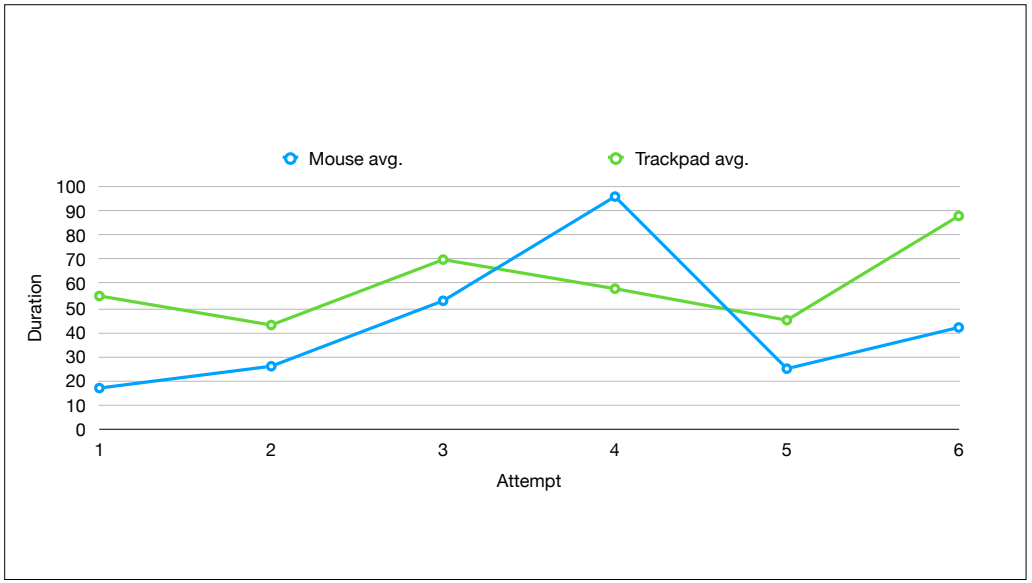


Graphs

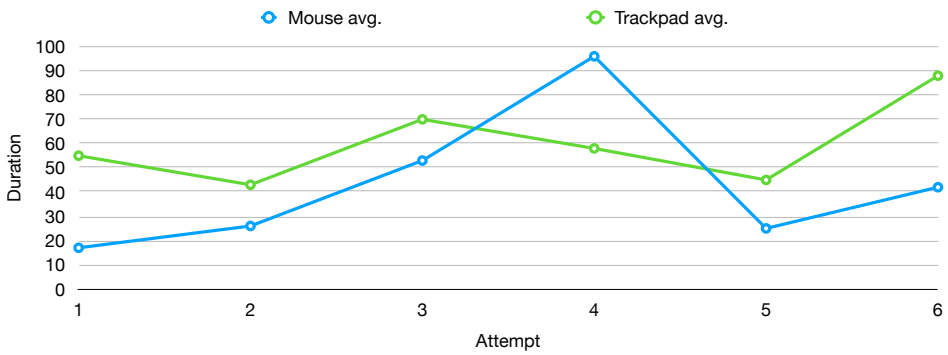


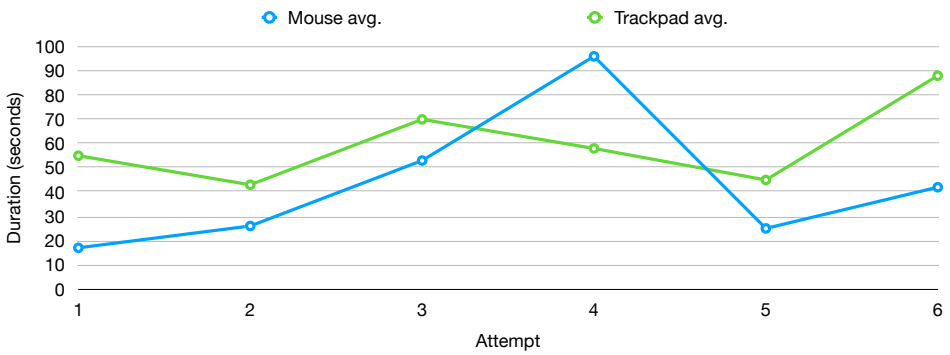

Mouse avg.
Trackpad avg.
100
90
80
70
60
50
40
30
20
10
0
1
2
3
4
5
6

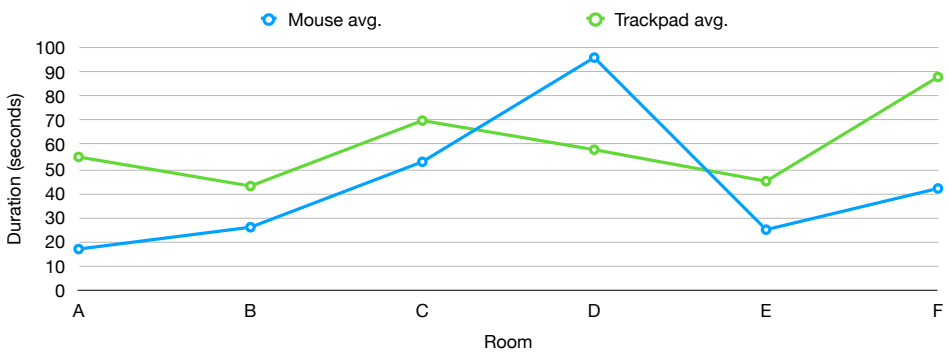

No axis labels
It could be whatever, label axis!
Mouse avg.
Trackpad avg.
100
90
80
70
60
50
40
30
20
10
0
1
2
3
4
5
6


Mouse avg.
Trackpad avg.
Duration
100
90
80
70
60
50
40
30
20
10
0
1
2
3
4
5
6
Attempt

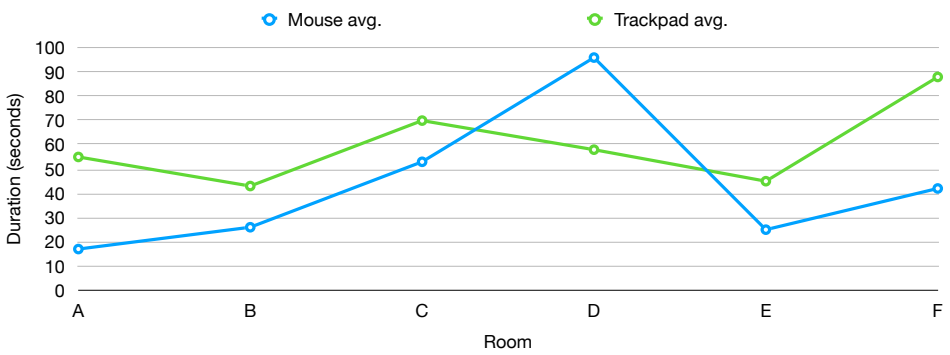
What's the unit of duration?

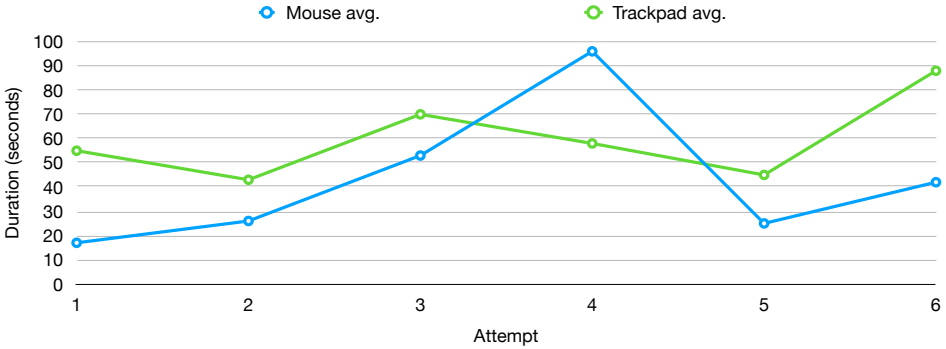


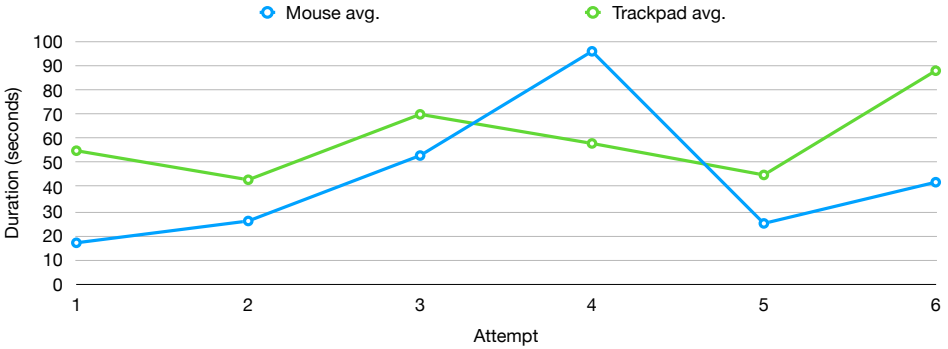


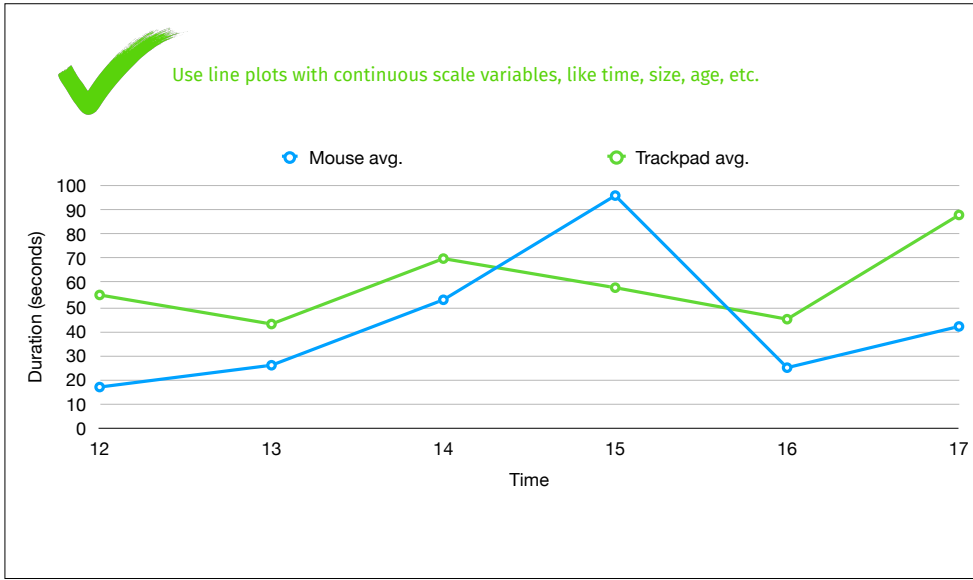


The room is a nominal variable, we have a discrete scale.
There is no meaning in the line between two "ticks".




Mouse avg.
Trackpad avg.
Duration (seconds)
100
90
80
70
60
50
40
30
20
10
0
1
2
3
4
5
6
Attempt


Still kind of wrong, since also attempts are a categorical variable.
Mouse avg.
Trackpad avg.
Duration (seconds)
100
90
80
70
60
50
40
30
20
10
0
1
2
3
4
5
6
Attempt


Use line plots with continuous scale variables, like time, size, age, etc.
Mouse avg.
Trackpad avg.
Duration (seconds)
100
90
80
70
60
50
40
30
20
10
0
12
13
14
15
16
17
Time


Precision

Name	Size
A	3212.18754127463
B	121.365978595142
C	12.5588485255459
D	9784.21894641212



A report is for human consumption, not for machines.

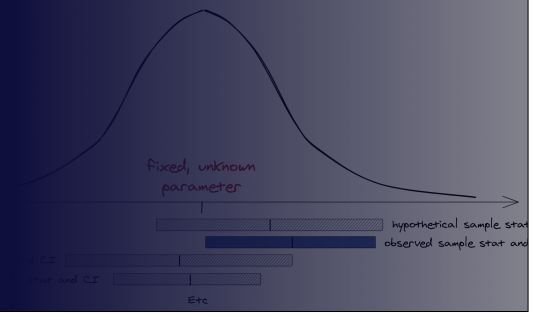
Name	Size
A	3212.18754127463
B	121.365978595142
C	12.5588485255459
D	9784.21894641212



Focus on the essential.

Name	Size
A	3212.2
B	121.4
C	12.6
D	9784.2

Confidence intervals



How certain can we be?

One observation

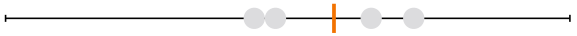
One observation

A sample

One observation

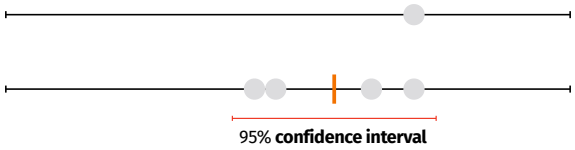
A sample

sample mean: point estimate



One observation

A sample
sample mean: point estimate

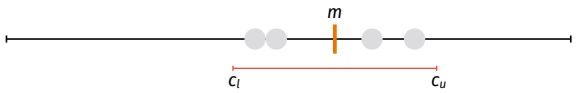


95% **confidence interval**

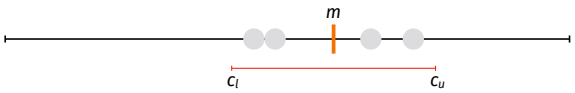
95% probability of **population** mean being in here

How to compute the confidence interval?

Computing the confidence interval



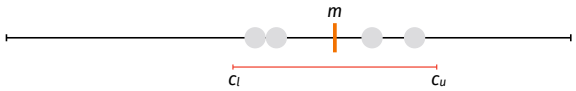
Computing the confidence interval



$$c_u = m + z_{a/2} \frac{S}{\sqrt{N}}$$

$$c_l = m - z_{a/2} \frac{S}{\sqrt{N}}$$

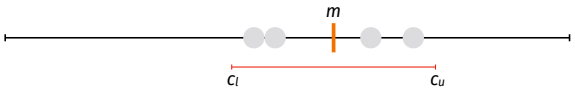
Computing the confidence interval



$$c_u = m + z_{a/2}\frac{S}{\sqrt{N}}$$

$$c_l = m - z_{a/2}\frac{S}{\sqrt{N}}$$

Computing the confidence interval

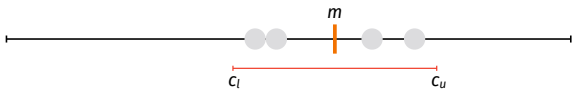


$$c_u = m + z_{a/2}\frac{S}{\sqrt{N}}$$

$$c_l = m - z_{a/2}\frac{S}{\sqrt{N}}$$

standard error of the mean

Computing the confidence interval



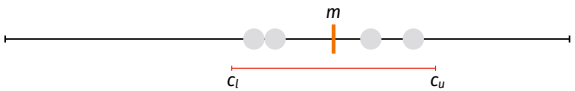
$$c_u = m + z_{a/2}\frac{S}{\sqrt{N}}$$

$$c_l = m - z_{a/2}\frac{S}{\sqrt{N}}$$

standard error of the mean

S is the standard deviation of the sample

Computing the confidence interval



$$c_u = m + z_{a/2}\frac{S}{\sqrt{N}}$$

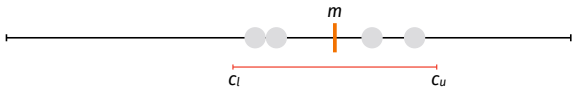
$$c_l = m - z_{a/2}\frac{S}{\sqrt{N}}$$

standard error of the mean

S is the standard deviation of the sample

N is the number of observations in our sample

Computing the confidence interval



$$c_u = m + z_{a/2}\frac{S}{\sqrt{N}}$$

$$c_l = m - z_{a/2}\frac{S}{\sqrt{N}}$$

significance level *a*

a the selected confidence interval, e.g., 0.95, 0.99

z see the z-table

https://www.statisticshowto.com/tables/z-table/

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936

Example

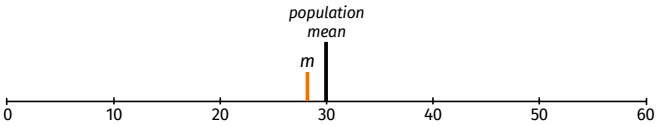
Population
Normally distributed, mean = 30

We will take one sample, having N = 30
30 observations

We compute the confidence interval
Confidence level: 90%

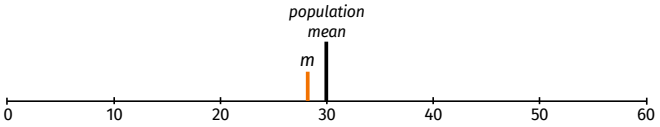
Sample

m = 28
S = 9
a = 0.90



Sample

$m = 28$
$S = 9$
$a = 0.90$

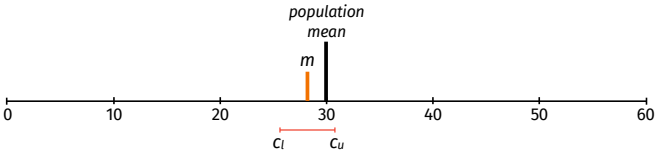


$$c_u = m + z_{a/2}\frac{S}{\sqrt{N}}$$

$$c_l = m - z_{a/2}\frac{S}{\sqrt{N}}$$

Sample

$m = 28$
$S = 9$
$a = 0.90$

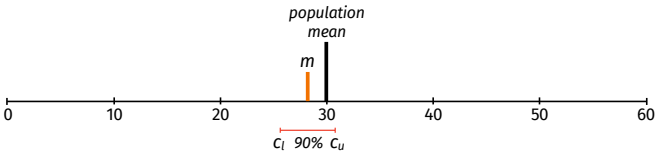


$$c_u = m + z_{a/2}\frac{S}{\sqrt{N}} = 28 + 1.64\frac{9}{\sqrt{30}} = 28 + 1.64 \text{ x } 1.64 = 30.69$$

$$c_l = m - z_{a/2}\frac{S}{\sqrt{N}} = 28 - 1.64\frac{9}{\sqrt{30}} = 28 - 1.64 \text{ x } 1.64 = 25.31$$

Sample

$m = 28$
$S = 9$
$a = 0.99$

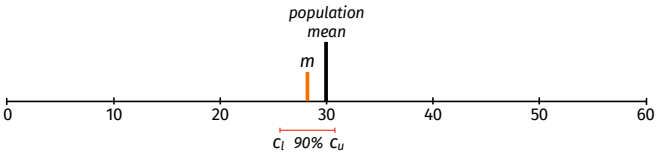


$$c_u = m + z_{a/2}\frac{S}{\sqrt{N}} = 28 + 1.64\frac{9}{\sqrt{30}} = 28 + 1.64 \text{ x } 1.64 = 30.69$$

$$c_l = m - z_{a/2}\frac{S}{\sqrt{N}} = 28 - 1.64\frac{9}{\sqrt{30}} = 28 - 1.64 \text{ x } 1.64 = 25.31$$

Sample

$m = 28$
$S = 9$
$a = 0.99$

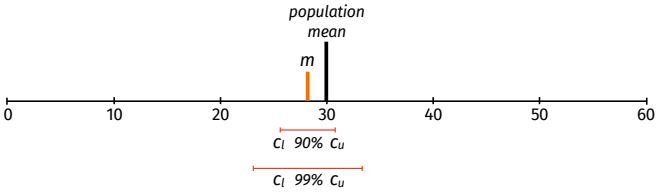


$$c_u = m + z_{a/2}\frac{S}{\sqrt{N}} = 28 + 2.57\frac{9}{\sqrt{30}} = 28 + 2.57 \text{ x } 1.64 = 30.69$$

$$c_l = m - z_{a/2}\frac{S}{\sqrt{N}} = 28 - 2.57\frac{9}{\sqrt{30}} = 28 - 2.57 \text{ x } 1.64 = 25.31$$

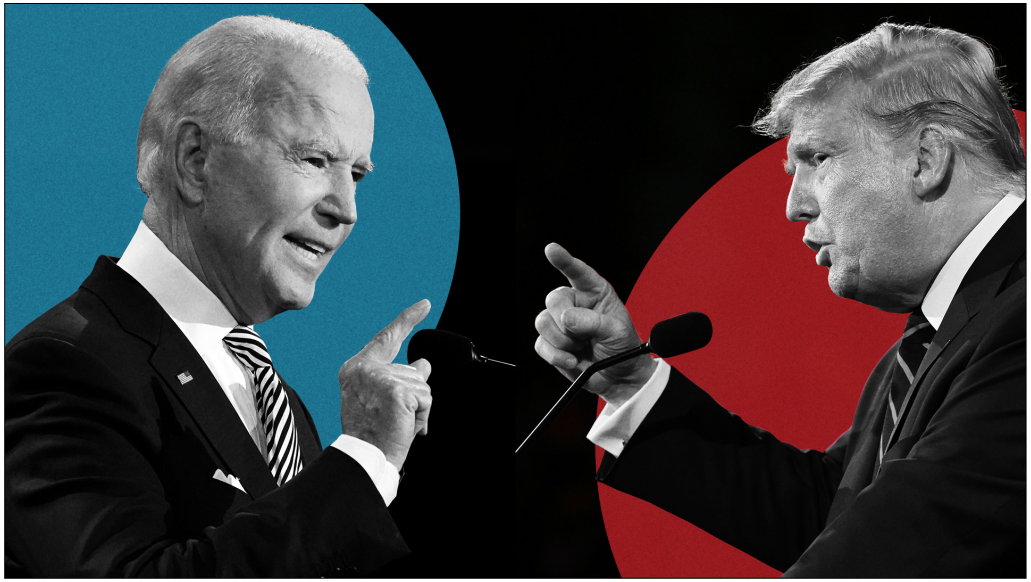
Sample

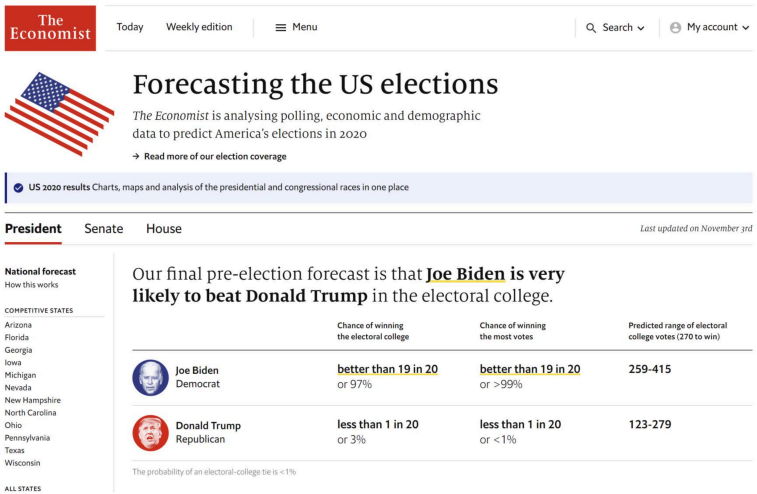
$m = 28$
$S = 9$
$a = 0.99$



$$c_u = m + z_{a/2}\frac{S}{\sqrt{N}} = 28 + 2.57\frac{9}{\sqrt{30}} = 28 + 2.57 \times 1.64 = 32.21$$

$$c_l = m - z_{a/2}\frac{S}{\sqrt{N}} = 28 - 2.57\frac{9}{\sqrt{30}} = 28 - 2.57 \times 1.64 = 23.79$$




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US 2020 results Charts, maps and analysis of the presidential and congressional races in one place

President Senate House

Last updated on November 3rd

National forecast
How this works

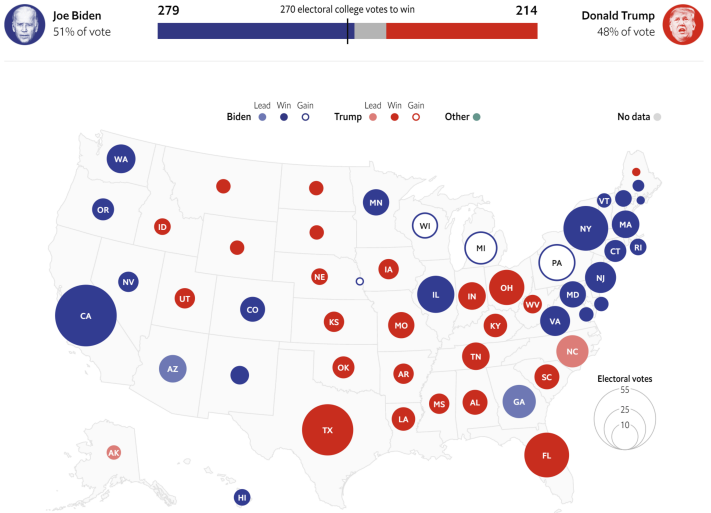
COMPETITIVE STATES
Arizona
Florida
Georgia
Iowa
Michigan
Nevada
New Hampshire
North Carolina
Ohio
Pennsylvania
Texas
Wisconsin

ALL STATES

Our final pre-election forecast is that **Joe Biden is very likely to beat Donald Trump** in the electoral college.

	Chance of winning the electoral college	Chance of winning the most votes	Predicted range of electoral college votes (270 to win)
Joe Biden Democrat	better than 19 in 20 or 97%	better than 19 in 20 or >99%	259-415
Donald Trump Republican	less than 1 in 20 or 3%	less than 1 in 20 or <1%	123-279

The probability of an electoral-college tie is <1%



The Economist

Our election model appears to have correctly predicted the outcome in 49 of 50 states and Washington, DC

